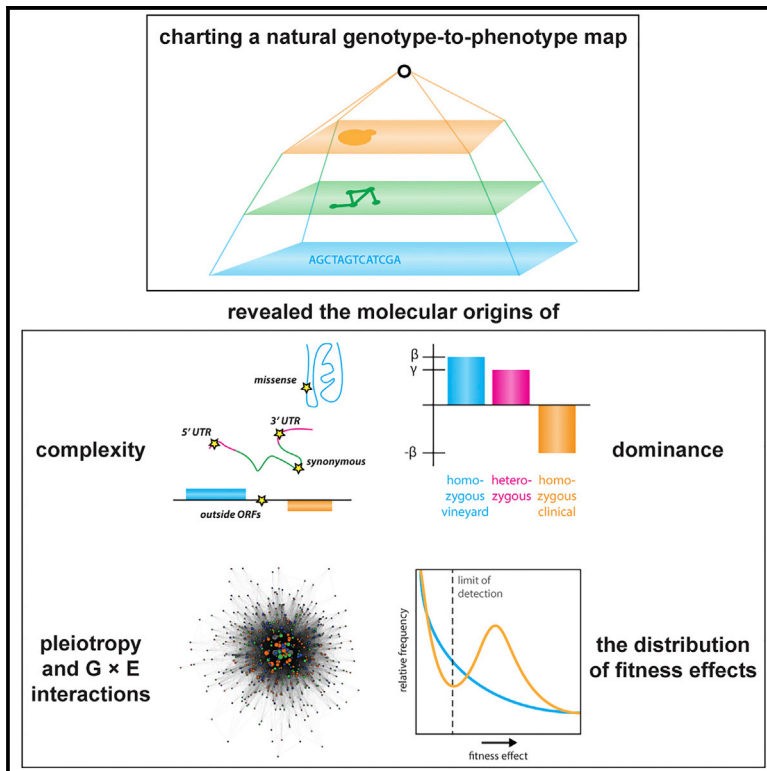


Molecular Origins of Complex Heritability in Natural Genotype-to-Phenotype Relationships

Graphical Abstract



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In Brief

The heritability of quantitative traits is intrinsically complex, but identifying its molecular origins is crucial for understanding how phenotypes emerge from genomes. Using a powerful genetic mapping approach, we discovered the molecular signatures of natural genetic variants that are important for phenotype. Many variants impact multiple traits, and their effects often switch between environments. Although they can be extremely complex, quantitative traits have finite linear contributors that can be comprehensively charted.

Highlights

- The genetic changes important to quantitative traits have diverse molecular signatures
- Key variants are broadly important, and their effects can change between environments
- Polymorphisms in disordered regions can have strong and widespread effects on phenotype
- Quantitative traits have finite numbers of linear contributors



Molecular Origins of Complex Heritability in Natural Genotype-to-Phenotype Relationships

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SUMMARY

The statistical complexity of heredity has long been evident, but its molecular origins remain elusive. To investigate, we charted 90 comprehensive genotype-to-phenotype maps in a large population of wild diploid yeast. In contrast to long-standing assumptions, all types of genetic variation contributed similarly to phenotype. Causal synonymous and regulatory variants exhibited distinct molecular signatures, as did nonlinearities in heterozygote fitness that likely contribute to hybrid vigor. Highly pleiotropic variants altered disordered sequences within signaling hubs, and their effects correlated across environments—even when antagonistic—suggesting that large fitness gains bring concomitant costs. Natural genetic networks defined by the causal loci differed from those determined by precise gene deletions or protein-protein interactions. Finally, we found that traits that would appear omnigenic in less powered studies do in fact have finite genetic determinants. Integrating these molecular principles will be crucial as genome reading and writing become routine in research, industry, and medicine.

INTRODUCTION

The intrinsic complexity of heritable traits has long been appreciated. Following the rediscovery of the work of Mendel more than a century ago, geneticists developed theories of heredity encompassing polygenicity, heterosis, and pleiotropy even before the molecular nature of the gene was understood (Fisher, 1919). Despite the evident statistical impact of these phenomena, we still lack a detailed molecular understanding of their origins. The idea that quantitative traits are driven by very large numbers of underlying loci has returned to prominence since the advent of practical whole-genome sequencing, but this “omnigenic” model remains largely untested (Boyle et al., 2017; Wray et al., 2018).

The budding yeast *Saccharomyces cerevisiae* has been a workhorse for establishing the architecture of heredity (Bloom et al., 2013; Costanzo et al., 2010) because targeted deletions and mapping studies using inbred crosses have much greater

power to detect small effects and second- and third-order interactions than genome-wide association studies (GWASs) in humans (Wu et al., 2017). However, studies in yeast and other models have been limited in important respects. First, the effect of natural genetic variation (e.g., missense variants) is seldom as dramatic as the deletion of an entire open reading frame (ORF) (Roy et al., 2018). Second, the haplotype blocks identified as causal often encompass many candidate variants. Lastly, most large-scale genetic mapping and deletion screening studies in yeast have been conducted in haploid strains, precluding the exploration of hybrid vigor and other diploid-specific phenomena.

To investigate the complexity of natural genotype-to-phenotype maps, we constructed a panel of 18,126 fully genotyped F₆ diploid *S. cerevisiae* segregants derived from wild yeast isolates. The scale of our experiments provided statistical power to discover quantitative trait loci (QTLs) of small effect, and the high meiotic crossover density in the segregants allowed us to resolve a substantial fraction of these QTLs to single causal nucleotides (quantitative trait nucleotides or QTNs). In contrast to deletion studies, our approach is highly sensitive to the wide range of effects on phenotype caused by the natural genetic variants. In all, we discovered 18,007 QTLs and 3,394 QTNs for 90 quantitative traits, implicating 1,644 of the 6,604 protein-coding genes in *S. cerevisiae*. This high-resolution atlas of heredity allowed us to define molecular mechanisms of polygenicity, heterosis, pleiotropy, and gene-environment interactions and to estimate the distribution of the fitness effects of extant genetic variants.

RESULTS

A Powerful, High-Resolution Genetic Mapping Panel

Typical genetic mapping panels in model organisms contain many more segregating markers than genotyped individuals (Bloom et al., 2013), with few exceptions (Ehrenreich et al., 2010); the same is true of GWASs of humans (Visscher et al., 2017). Moreover, marker variants are usually in strong linkage disequilibrium with other nearby variants, and when microarrays are used for genotyping, these haplotype blocks may also contain additional, unidentified polymorphisms (Schaid et al., 2018). Furthermore, causal variants may be rare, reducing their statistical effect on the mapping population as a whole (Gibson, 2012).

To address these shortcomings, we generated a panel of 18,126 fully genotyped F₆ diploid progeny of a cross between a pathogenic *S. cerevisiae* isolate, YJM975, from an immunocompromised patient in Italy (McCullough et al., 1998) and a Zinfandel



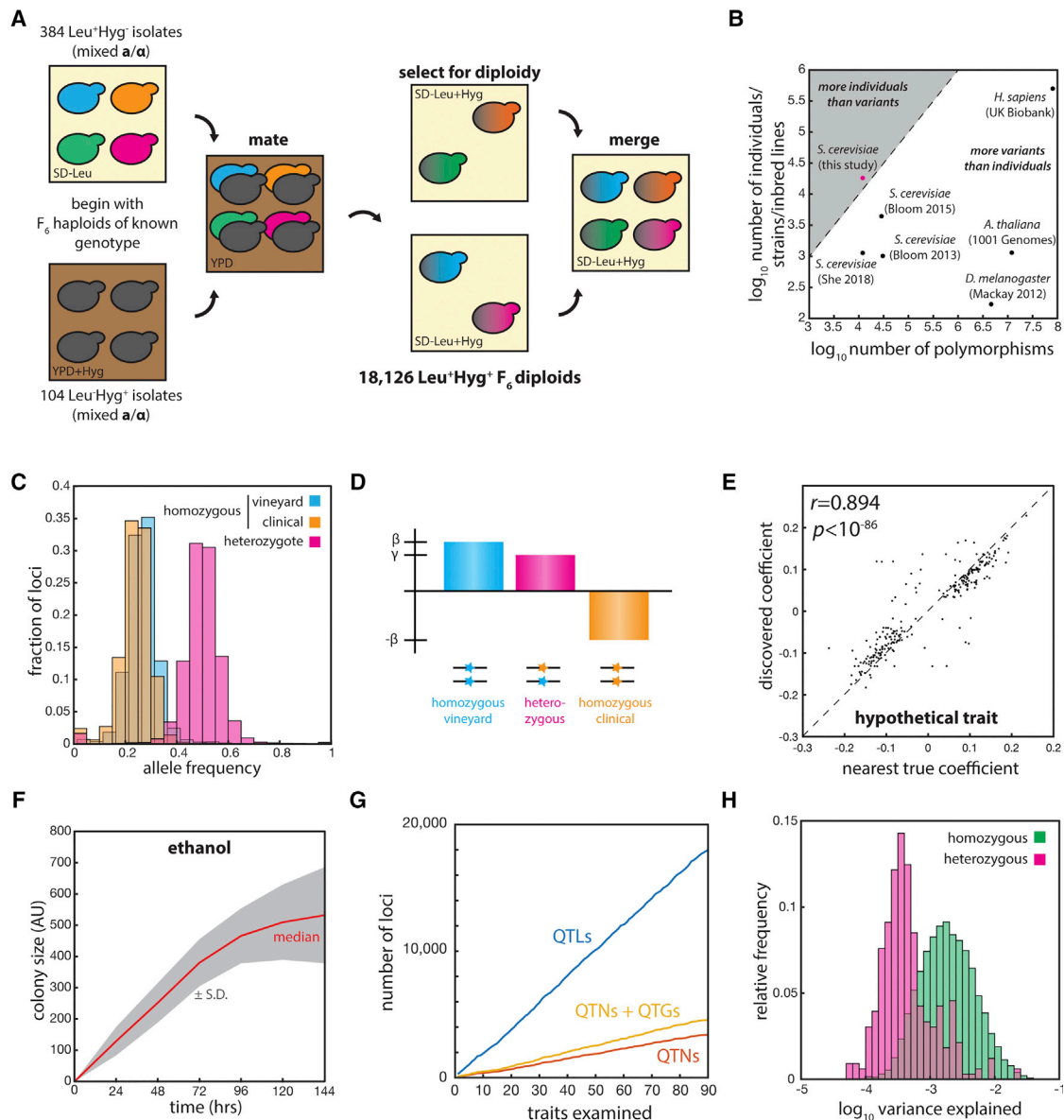


Figure 1. An Extremely Large Panel of Fully Genotyped Diploid Yeast to Inventory Complex Heredity

(A) Mating scheme used to construct the diploid segregant collection.

(B) Number of segregating genetic variants and number of genotyped individuals in various mapping panels.

(C) Relative frequencies of RM11 homozygotes (blue), YJM975 homozygotes (orange), and heterozygotes (magenta) across the 12,054 polymorphic loci in the panel.

(D) Scheme illustrating the linear mixed model used to describe phenotype. β represents the homozygous locus effect and γ the heterozygous locus effect, if any.

(E) Correlation (Pearson's r) between effect size in the model and nearest true locus effect for an example simulated trait; p value by Student's t distribution.

(F) Growth of segregant panel on S-CSM + 2% ethanol solid medium.

(G) Total number of QTLs, QTGs, and QTNs discovered with increasing number of quantitative traits examined.

(H) Histogram of variance explained per QTL for linear (homozygous, green) and nonlinear (heterozygous, magenta) contributions across all 90 quantitative traits. See also Figure S1.

grape isolate, RM11-1a, from a California vineyard (Török et al., 1996) (Figure 1A). In contrast to most previous mapping panels in model organisms, our population contains more genotyped individuals than segregating genetic variants (Figure 1B). The use of inbred *S. cerevisiae* strains also results in highly uniform minor allele frequencies, allowing equivalent sensitivity to the effects of all segregating polymorphisms (Li et al., 2017) (Fig-

ure 1C). Importantly, the segregating variants are in very low linkage disequilibrium, allowing *in silico* allele swaps to identify causal variants with single-nucleotide resolution (She and Jarosz, 2018). We first mapped the linear contributions of homozygous loci and then considered nonlinearities attributable to partial or overdominance by allowing heterozygous loci to adopt coefficients that deviated from the homozygous midpoints (Figure 1D).

To characterize the performance of our mapping panel and analysis procedure, we performed rigorous simulations using 50 highly complex ground-truth genetic architectures. Each hypothetical trait comprised $n = 275$ underlying causal loci of random effect and sign, with realistic levels of Gaussian noise based on the broad-sense heritability of the traits we mapped. Comparison of *in silico* mapping of the simulated phenotypes to the known architectures revealed that our regression approach recalled $94\% \pm 2.8\%$ of true underlying QTL and identified true underlying QTN with $92\% \pm 7.1\%$ precision ($n = 50$ hypothetical traits; mean $\pm$ SD) (Figure S1A). Moreover, our mapping procedure accurately captured the effect size of the true causal loci for a range of underlying effect sizes (Pearson's $r = 0.93 \pm 0.03$ for $n = 50$ traits; mean $\pm$ SD) (Figure S1A, e.g., Figure 1E). These comprehensive tests confirmed that our panel was powered to dissect highly complex traits into their constituent loci, accurately identify effect sizes, and identify the genes or causal variants associated with each QTL. We were able to not only identify nearly all linear contributors to phenotype but also to resolve them to single genes and, in many cases, single nucleotides.

Diverse Contributors to Complex Traits

We next phenotyped the segregant panel in 15 environmental conditions (including various carbon sources and toxins, an FDA-approved drug, and other stresses) across six time points (e.g., Figure 1F). We considered each time point in each environment as a separate quantitative trait, and the $\sim 1,600,000$ growth measurements allowed us to identify 18,007 QTLs at an empirical false-discovery rate of $1.5\% \pm 2.1\%$ (by permutation test; mean $\pm$ SD), with 200 ± 52.2 QTLs identified per trait (mean $\pm$ SD) (Figure S1B). Our model explained $72.8\% \pm 18.5\%$ of the broad sense heritability across the 90 traits examined (mean $\pm$ SD) (Figure S1C), and we readily discovered loci explaining as little as 0.01% of phenotypic variance (Figures S1D and S1E). The remaining “missing heritability” is likely due to second- or higher-order genetic interactions (Bloom et al., 2015; Poelwijk et al., 2017). Most phenotypic variance was explained by linear homozygous contributions ($n = 3,165$), but numerous heterozygous contributions (of $n = 229$ total) had effect sizes comparable to homozygous terms (Figure 1H).

Of the QTLs identified, we unambiguously mapped 3,394 with single-nucleotide resolution, corresponding to 1,608 unique causal variants. An additional 1,166 QTLs could be resolved to a single quantitative trait gene (QTG) (Figure 1G). Strikingly, fully 24.9% of the 6,604 protein-coding genes (and 13.3% of the individual segregating polymorphisms) were implicated in determining growth across this comparatively small number of environments. Thus, it is possible that most segregating variants have the potential to significantly contribute to phenotype in the highly complex, varied environments faced by *S. cerevisiae* in nature (Jakobson et al., 2019).

Molecular Mechanisms of Coding, Non-coding, and Extragenic Causal Variants

The large number of QTNs we identified allowed us to examine diverse molecular contributions to heredity (Figures 2A and 2B). Missense variants exhibited the greatest variance explained, followed closely by synonymous and extragenic vari-

ants (Figure 2C). These same trends were reflected in the regression coefficients, which better represent the impact of the alleles in each segregant (Figures 2D and S1F). Many different amino acid substitutions were represented in the pool of causal missense variants (Figure 2E), and their effect sizes correlated with molecular expectation: substitutions with lower BLOSUM62 scores (i.e., more perturbative amino acid substitutions) were of larger effect (Pearson's $r = -0.201$, $p < 0.04$) (Figure 2F). Despite this, the effect of each variant remained context dependent, and BLOSUM62 scores were only modestly predictive of the effect sizes, underscoring the importance of explicitly assessing the effects of coding variants (Diss and Lehner, 2018; Fowler and Fields, 2014). A recently developed method based on deep learning of saturating mutagenesis data (Gray et al., 2018) was less predictive than BLOSUM62, perhaps because natural missense variants are conservative relative to the broad spectrum of variants explored in saturation mutagenesis studies (Figure 2G).

Although missense variants had the largest effect on phenotype ($p < 0.001$ by two-sample Kolmogorov-Smirnov test), the effect-size distributions of all variant classes overlapped (Figure 2C). Synonymous natural variants, often regarded as unlikely to significantly affect phenotype (Kumar et al., 2009), had median effect sizes that were comparable to those of missense variants and larger than those of extragenic variants ($p < 10^{-6}$ by two-sample Kolmogorov-Smirnov test). To probe the molecular origin of this relationship, we assessed the relative positions of causal synonymous variants within a genome-wide meta-ORF relative to all synonymous variants segregating in the cross (Figure 2H). Causal variants were strongly enriched at the 5' end of the meta-ORF relative to all synonymous variants ($p < 0.0003$ by two-sample Kolmogorov-Smirnov test). Moreover, synonymous causal variants at the 5' end of the meta-ORF exhibited larger changes in codon adaptation index (CAI) than all segregating synonymous variants ($p < 0.01$ by two-sample Kolmogorov-Smirnov test) (Figures 2I and 2J) (Drummond et al., 2006). There was no correlation between the sign of the change in CAI and the sign of the effect on phenotype, likely because both increased and decreased translation could improve growth, depending on the gene product. Our data are consistent with a model in which effects due to synonymous codons are most pronounced in the early stages of translation and folding, possibly as part of an adaptive “translation ramp” that favors slow translation at the beginning of genes (Tuller et al., 2010).

Next, we examined causal variants lying outside annotated ORFs. The positions of these causal variants were not distinct from all segregating extragenic variants relative to the positions of annotated transcriptional start (TSS) and end (TES) sites (Figures S2A and S2B). However, when we examined predicted transcription-factor occupancy (Pachkov et al., 2007), we found that extragenic causal variants were enriched at sites of both high and low transcription-factor binding (Figure S2C). That is, these variants may act either by changing transcription-factor affinity or by influencing genome structure and accessibility independent of transcription-factor-binding sites. The former could directly precipitate changes in transcriptional regulation, consistent with the observation of abundant *cis*-eQTLs in *S. cerevisiae* (Kita et al., 2017). The latter may be attributable to perturbations in poly(dA:dT) tracts that are involved in nucleosome

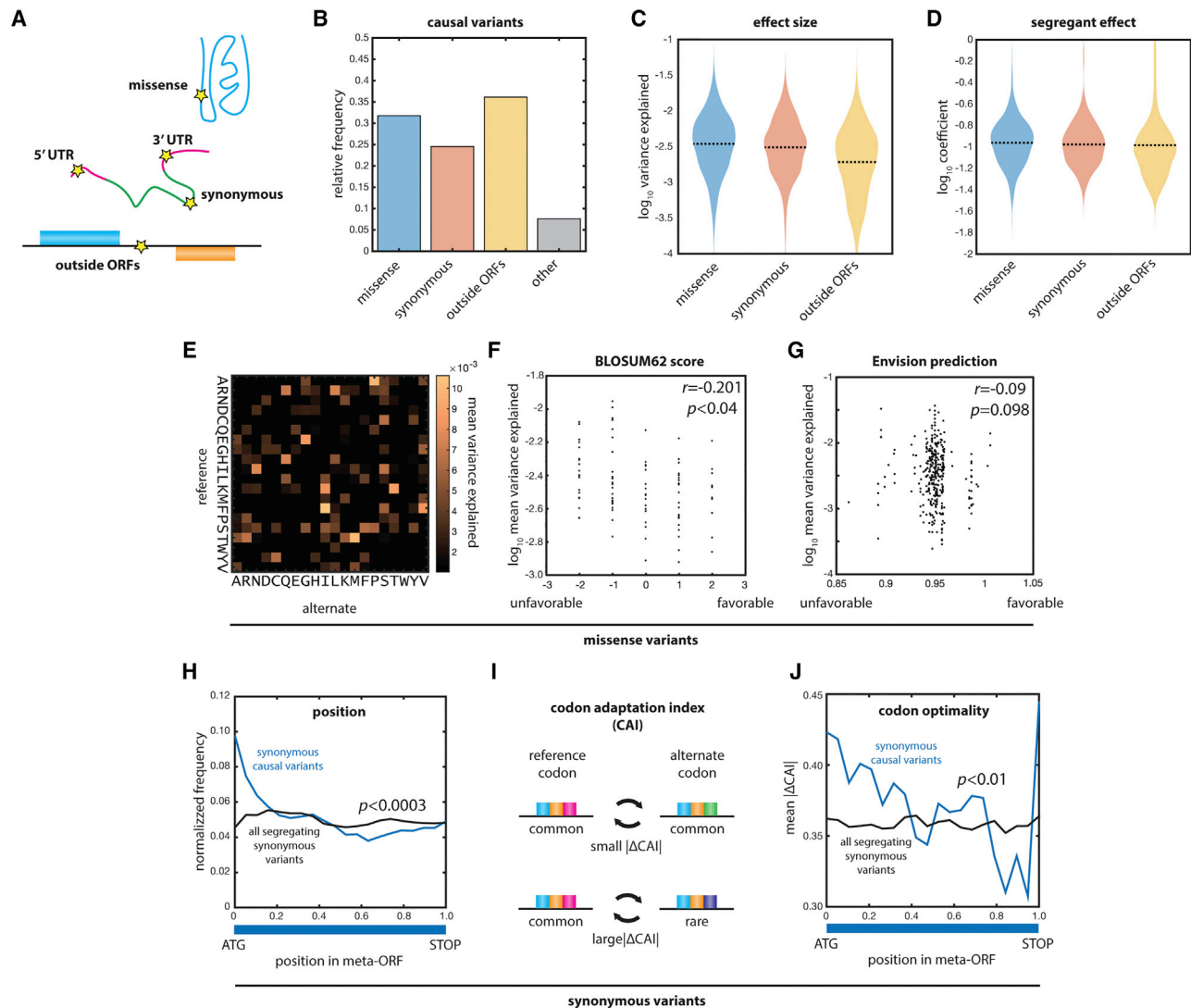


Figure 2. Diverse Molecular Mechanisms Underlie Genetic Complexity

(A) Classes of molecular variation responsible for phenotype.

(B) Relative frequencies of all types of causal variants.

(C) Variance explained by missense, synonymous, and extragenic causal variants.

(D) Effect size of missense, synonymous, and extragenic causal variants.

(E) Mean variance explained by missense variants (ordinate, reference residue; abscissa, alternate residue).

(F) Mean variance explained as a function of BLOSUM62 score for missense variants. Correlation by Pearson's r ; p value by Student's t distribution.

(G) Mean variance explained as a function of Envision score for missense variants. Correlation by Pearson's r ; p value by Student's t distribution.

(H) Position of synonymous causal variants (blue) and all segregating synonymous variants (black) as a function of position within the meta-ORF (ATG, start codon; TAA, stop codon). p value by Kolmogorov-Smirnov test.

(I) Schematic of large and small changes in codon optimality.

(J) Mean absolute change in codon adaptation index (CAI) as a function of position within the meta-ORF. p value by Kolmogorov-Smirnov test.

See also Figure S1.

organization (Segal and Widom, 2009). Indeed, our analysis of the local sequence context (± 5 nt) of these causal variants revealed striking clusters of polyA and polyT motifs (Figure 3A). More than 77% of extragenic causal variants were in contexts with greater than 50% A/T content, and 23.6% were in contexts with greater than 75% A/T content.

Reasoning that biophysical perturbations to the genome might impact chromatin structure and thus transcriptional regu-

lation, we next examined the histone marks near extragenic causal variants (Weiner et al., 2015). We found enrichments for many histone marks associated with open, active chromatin in the vicinity of causal variants, including H3K14ac, H3K18ac, H3K4me3, and H3K9ac (Figure 3B), and the prevalence of many of these marks near causal variants was significantly correlated with effect size (Figure S2D). Together, these data indicated that extragenic variants in open chromatin adjacent

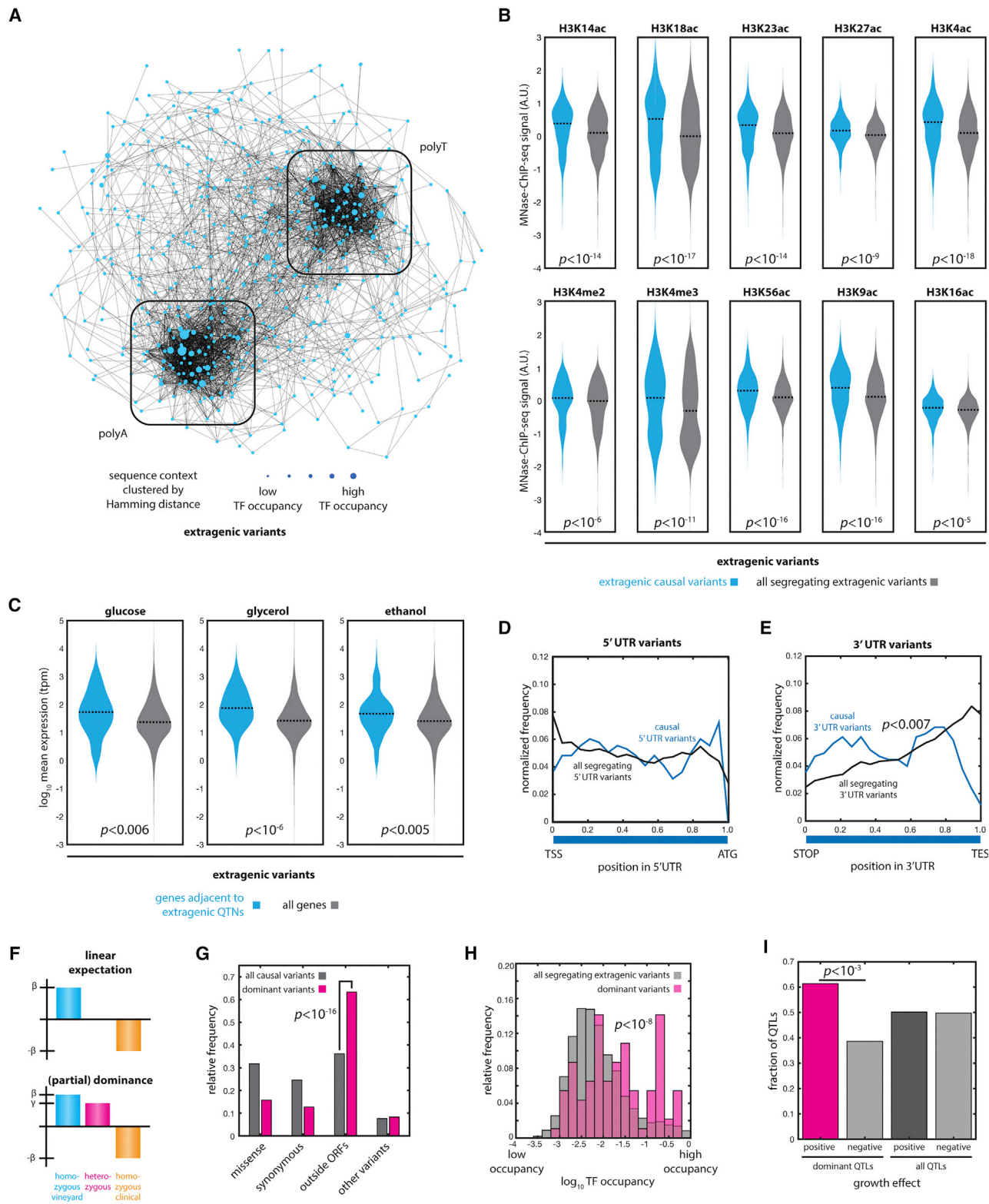


Figure 3. Molecular Signatures of Causal Regulatory Variation

(A) Spring-embedded network representation of local sequence context of extragenic causal variants, weighted by Hamming distance. Boxes indicate polyA- and polyT-enriched clusters. Nodes are sized by predicted transcription factor occupancy.

(B) Mean MNase-ChIP-seq signal for the indicated histone marks for nucleosomes within 200 nt of causal extragenic variants (blue) and all segregating extragenic variants (gray). Bonferroni-corrected p values by Kolmogorov-Smirnov test.

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to actively transcribed genes were more likely to impact phenotype.

To test this hypothesis, we measured genome-wide mRNA levels by RNA sequencing (RNA-seq) in the diploid parental strains (RM11a/ α and YJM975a/ α) during exponential-phase growth in several conditions (2% glucose, 2% glycerol, and 2% ethanol; [Data S2](#)). The mean expression (averaged across both parental genetic backgrounds) was significantly higher for genes adjacent to identified extragenic QTNs in that environment as compared to all genes ($p < 0.006$; $p < 10^{-6}$; $p < 0.005$, respectively, by two-sample Kolmogorov-Smirnov test) ([Figure 3C](#)). Extragenic QTNs were also adjacent to genes with higher variability in expression, although this effect was driven primarily by the correlation between mean expression and expression variability ([Figure S2E](#)). In contrast, no enrichment for increased expression was observed for genes containing missense QTNs ([Figure S2F](#)), indicating that extragenic QTNs were indeed playing a preferential role in controlling the expression of actively transcribed genes. The fold change in expression of a gene between the homozygotes was not predictive of QTG status, perhaps because the expression levels in the parents represent a complex integration of both *cis* and *trans* regulation ([Wittkopp et al., 2004](#)).

Many causal variants arose in the 5' and 3' UTRs of ORFs. Causal variants in the 5' UTRs exhibited no significant spatial enrichment ([Figure 3D](#)), whereas those in 3' UTRs were markedly enriched at the 5' (ORF-proximal) end of the 3' UTR and depleted near the end of the transcript relative to all segregating 3' UTR variants, suggesting a role for these variants in translation termination ($p < 0.007$ by two-sample Kolmogorov-Smirnov test) ([Figure 3E](#)). The nucleotides immediately following the STOP codon are known to impact the efficiency of translation termination ([Namy et al., 2001](#)), and other 3' UTR variants may impact the stability of the mRNA transcript as a whole ([Shalgi et al., 2005](#)).

Finally, we used a sign test ([Fraser et al., 2010](#)) to search for a signature of lineage-specific selection across the many linear contributors to phenotype that we observed. All of the causal variants we identified likely have a selection coefficient greater than $\sim 1/N_e$, the threshold for the action of selection (which is very small, $\sim 10^{-5}$ – 10^{-6} , for organisms such as fungi). Therefore, one would expect to observe a coherent signature of adaptation as evidenced by spatial clusters of variants from one parent with the same effect on phenotype, as was recently observed in *S. cerevisiae* for other causal variants ([Sharon et al., 2018](#)). Indeed, we observed that nearby pairs of variants from the same parent were significantly more likely than would be expected by chance to have the same effect on phenotype ([Figure S2G](#)), even over long genomic distances. This observa-

tion suggests that selection has acted coherently on variants of widely varying effect size ([Jakobson et al., 2019](#)), perhaps as a consequence of the recent adaptation of RM11 and YJM975 to their fermentation and human-host-associated niches, respectively.

Dominance Loci Disrupt Regulation

Heterosis, also called hybrid vigor, is the tendency for hybrids to outperform their parents. The phenomenon is widespread in organisms from yeast to agricultural crops, yet our understanding of its molecular origins is limited to a few individual cases ([Chen, 2013](#)). In addition to diverse linear contributors to phenotype, we identified extensive nonlinearities in the behavior of heterozygotes ([Figure 3F](#)). We refer to these loci collectively as dominance QTNs, encompassing partial, under-, and over-dominance. The dominance QTNs we discovered ($n = 229$) were strongly enriched with extragenic (presumably regulatory) variants relative to all causal variants we identified ($p < 10^{-16}$; Fisher's exact test) ([Figure 3G](#)). Moreover, dominance QTNs were enriched in regions of high transcription-factor occupancy relative to all segregating extragenic variants ($p < 10^{-8}$ by two-sample Kolmogorov-Smirnov test) ([Figure 3H](#)). Lastly, the coefficients for dominance loci were predominantly positive, i.e., heterozygotes typically exhibited greater growth than would be expected from a linear model (178 positive coefficients of 290 total; 61.4%; $p < 10^{-3}$ by Fisher's exact test) ([Figure 3I](#)). Theory predicts that this skew in heterozygote fitness should naturally result from adaptation in diploid populations ([Sellis et al., 2011](#)). Together, these observations suggest a molecular model of dominance in which changes in regulatory interactions disrupt processes that would ordinarily limit growth under stress ([Bar-Zvi et al., 2017](#)), driving hybrid vigor.

Heterozygotes of the Rds1 transcription factor ($RDS1^{\text{Gln695/Lys695}}$), for instance, exhibited improved growth in 2% galactose relative to the expectation based on a linear model. The polymorphic residue in this protein, Gln695, is located near the C terminus, distal to the N-terminal DNA-binding region. A neighboring homozygous variant, $Rds1^{\text{Ser352Asn}}$, affected growth in 2% ethanol, 2% glycerol, 2% raffinose, and 2% maltose at 37°C, suggesting that the Rds1 regulatory hub may play a role in adaptation to many environments despite the low copy number of the Rds1 protein when cells are grown in rich medium ([Kulak et al., 2014](#)). Indeed, genes observed to be upregulated by Rds1 form a highly connected network (protein-protein interaction $p < 10^{-16}$; STRING database) that is enriched with genes involved in "starch and sucrose metabolism" ($p < 0.001$, STRING database). The reported targets include *YGP1*, *SPI1*, and *GLK1*, all of which are differentially

(C) Mean expression (TPM) across both parental diploids ($n = 3$ biological replicates per strain) for genes adjacent to extragenic QTN and all genes, measured during growth in media containing 2% glucose, 2% glycerol, and 2% ethanol, as indicated. p values by Kolmogorov-Smirnov test.

(D) Position of all causal variants in the 5' UTR (blue) and all segregating variants in the 5' UTR (black) as a function of position within the pseudo-UTR (TSS, transcription start site; ATG, start codon). p value by Kolmogorov-Smirnov test.

(E) Position of all causal variants in the 3' UTR (blue) and all segregating variants in the 3' UTR (black) as a function of position within the pseudo-UTR (TAA, stop codon; TES, transcription end site). p value by Kolmogorov-Smirnov test.

(F) Impact of dominance in our linear mixed model of phenotype; shown is an example of partial dominance.

(G) Variant types of all causal variants (gray) and dominant causal variants (magenta). p value by Fisher's exact test.

(H) Transcription factor occupancy at positions of all extragenic causal variants (gray) and dominant variants (magenta). p value by Kolmogorov-Smirnov test.

(I) Number of dominant QTLs with a positive (magenta) or negative (gray) effect as compared to all QTLs. p value by Fisher's exact test.

See also [Figure S2](#).

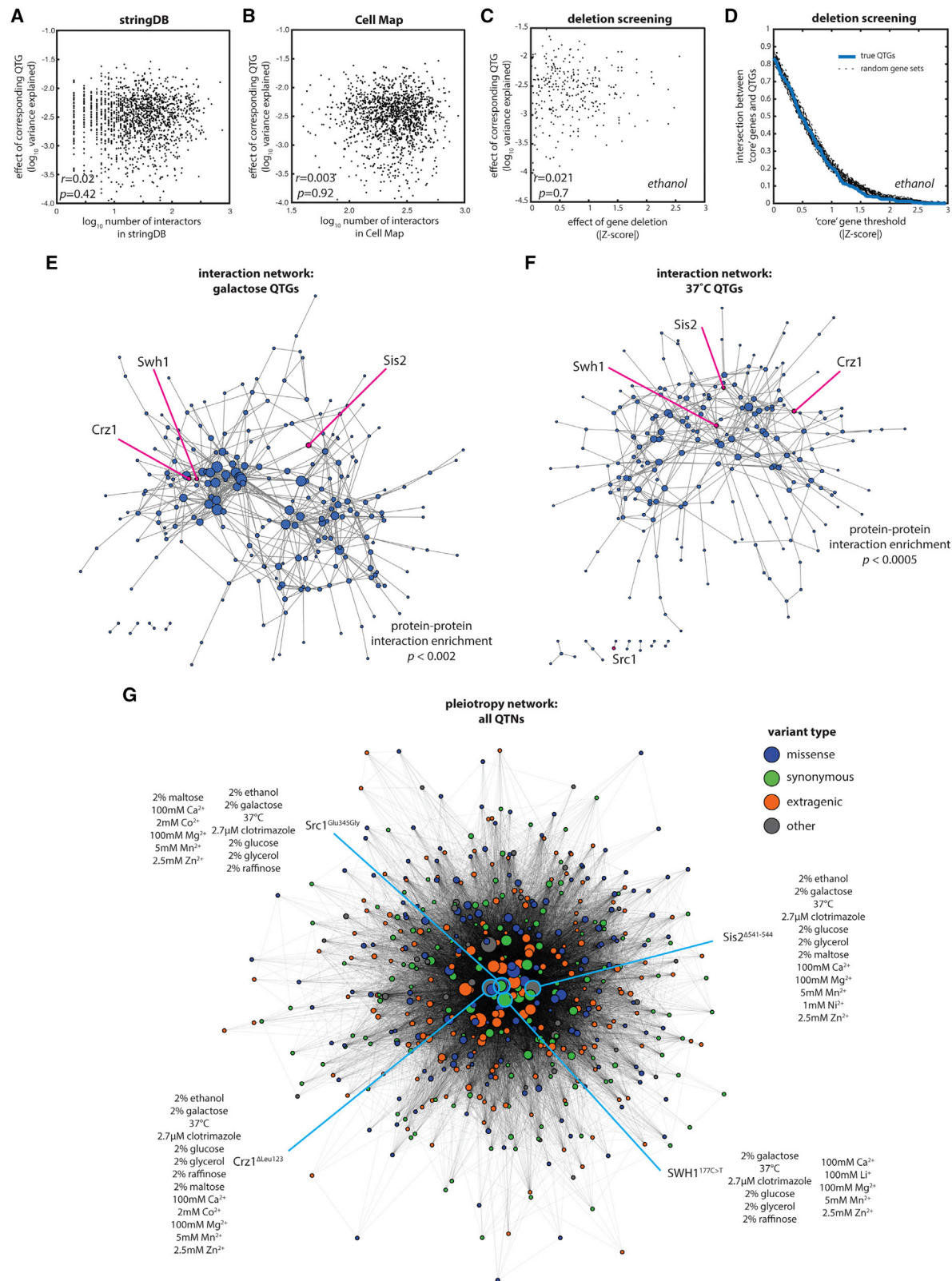


Figure 4. The Variant-to-Phenotype Map Is Distinct from Other Maps of Cellular Connectivity

(A–C) QTG effect size as a function of (A) connectivity in STRING database, (B) connectivity in the Cell Map, and (C) gene deletion or DAMP allele effect. Correlations by Pearson's r ; p values by Student's t distribution.

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regulated in response to stress- and metabolism-related reprogramming (Stanley et al., 2010).

Two other non-coding variants exhibiting dominance also seemed likely to affect gene regulation: a TTTT deletion at position 134,112 of chromosome X, lying between *INO1* and *VPS35*, and a T insertion at position 409,806 of chromosome VII, lying between *TIF463* (eIF4G) and *RPT6*. Both variants are in poly(dA:dT) tracts, which are associated with transcriptional regulation and, as noted above, are enriched for extragenic causal variants (Yagil, 2006). These observations are consistent with a model of heterosis in which a single copy of a polymorphism, whether in a diffusible factor or a regulatory region, can be sufficient to meaningfully alter a gene control program.

Natural Causal Variants Form Coherent Biological Networks Distinct from Those Defined by Whole-Genome Deletions

In yeast, most previous efforts to identify causal genes and the interactions between them have focused on precise ORF deletions (Costanzo et al., 2010, 2016; Giaever et al., 2002). We tested whether the topology of the variant-to-phenotype mapping we discovered was similar to those determined previously. We first compared the effect size of each identified QTG to the number of total genetic and physical interactors with that gene in the STRING database (derived from known and predicted protein-protein and genetic interactions) (Figure 4A) (Szkarczyk et al., 2017). We also compared the effect size of each QTG to the number of significant genetic interactors of the gene in the Cell Map database (derived from genetic interaction scores in precise double-deletion backgrounds) (Figure 4B) (Costanzo et al., 2016). The connectivity of the gene was not predictive of effect size in either case. Finally, because these databases are based primarily on data collected in rich medium and to control for growth assay format, we phenotyped the *S. cerevisiae* deletion and DAmP (hypomorphic allele) collections (Breslow et al., 2008; Giaever et al., 2002) in 2% ethanol under the same growth conditions used in our mapping experiment. Even in this case, the effect size of QTGs in 2% ethanol was not significantly correlated with the strength of its corresponding deletion or DAmP allele phenotype (Figure 4C). Although we and others have previously found that some genes are identified as important in both deletion and mapping studies (She and Jarosz, 2018), our observations suggest that the genotype-to-phenotype map for natural genetic variation is fundamentally topologically distinct from that derived from gene deletions.

One method to define the “core” genes important for a trait in yeast is to measure the phenotype of all precise ORF deletions. We defined core genes for growth on 2% ethanol by using a sliding window based on the effect size we measured for the precise gene deletions (Data S3). At no effect size threshold were core genes defined in this manner enriched with QTGs relative to randomly selected sets of genes (Figure 4D), although the net-

works of QTGs we identified were in some cases enriched with protein-protein interactions (Figures 4E and 4F). Notably, however, causal variants that impacted large numbers of quantitative traits (highlighted and discussed in more detail below) were not central to these networks, again suggesting that the network of phenotypic connectivity between genetic variants (Figure 4G) is distinct from the molecular networks defined by genetic or physical interactions. Together, these results suggest that gene hits from deletion screens should not be the only loci at which to search for consequential natural genetic variation.

Abundant Synergy and Antagonism in Pleiotropy and Gene-Environment Interactions

Pleiotropy and gene-environment interactions impact the topology of an organism’s fitness landscape by embedding synergies and tradeoffs in the effects of genetic changes on fitness (Wagner and Zhang, 2011). Many detrimental alleles in humans are thought to have been maintained because of antagonistic pleiotropy (Corbett et al., 2018), and theory predicts that the fate of newly arising variants is strongly influenced by their varying effects across environments (Pavličev and Cheverud, 2015). Yet, these effects have remained difficult to assess because mapping intervals typically contain multiple candidate causal variants (Solovieff et al., 2013). Our discovery of large numbers of unambiguous causal variants for numerous quantitative traits allowed us to systematically assess the prevalence of synergistic and antagonistic interactions in the genotype-to-phenotype map of our segregant panel (Figure 5A).

The growth of a segregant was generally correlated across traits, and this growth correlation (in terms of Pearson’s *r*) was itself correlated with the extent of QTN overlap (a proxy for pleiotropy and gene-environment interactions) (Figures S3A and S3B). This suggested that such interactions were common and predominantly synergistic, resulting in macroscopic phenotypic correlation. This was indeed the case at the molecular level: 39.5% of QTNs influenced more than one of the quantitative traits we examined (Figure 5B). Moreover, 34% of QTNs influenced growth in more than one distinct environment (Figure 5C).

The vast majority (97.7%) of pleiotropic interactions within a given environmental condition were synergistic across time points, suggesting that pronounced tradeoffs between growth in the lag and exponential growth phases, for instance, were uncommon (Figures 5D and 5J). Since the genetic architectures of traits measured at successive time points are not independent, we separately assessed gene-environment interactions occurring between traits measured independently. In this case, by contrast, a substantial fraction (12.8%) of gene-environment interactions were antagonistic (Figures 5F and 5J). These antagonistic interactions imply significant intrinsic phenotypic tradeoffs, even among the relatively small array of conditions we tested. The instances we identified involved diverse biological processes, including glycosylation and ergosterol biosynthesis,

(D) Enrichment of QTGs in core gene sets as defined by a sliding window of absolute *Z* score of gene deletion or DAmP allele effect. Dashed lines show the same enrichment calculated for random gene sets of the same size.

(E and F) Spring-embedded protein-protein interaction maps from STRING database for QTG in (E) galactose and (F) 37°C with edges weighted by interaction strength. Nodes are sized by interaction degree. Highlighted are key, highly pleiotropic genes. *p* values by Student’s *t* distribution.

(G) Spring-embedded network representation of QTNs (nodes) connected by edges weighted by the number of traits in which the variants are jointly causal and colored by the type of molecular variation as indicated. Nodes sized by extent of pleiotropy of the QTN. Four key pleiotropic QTNs are highlighted in blue.

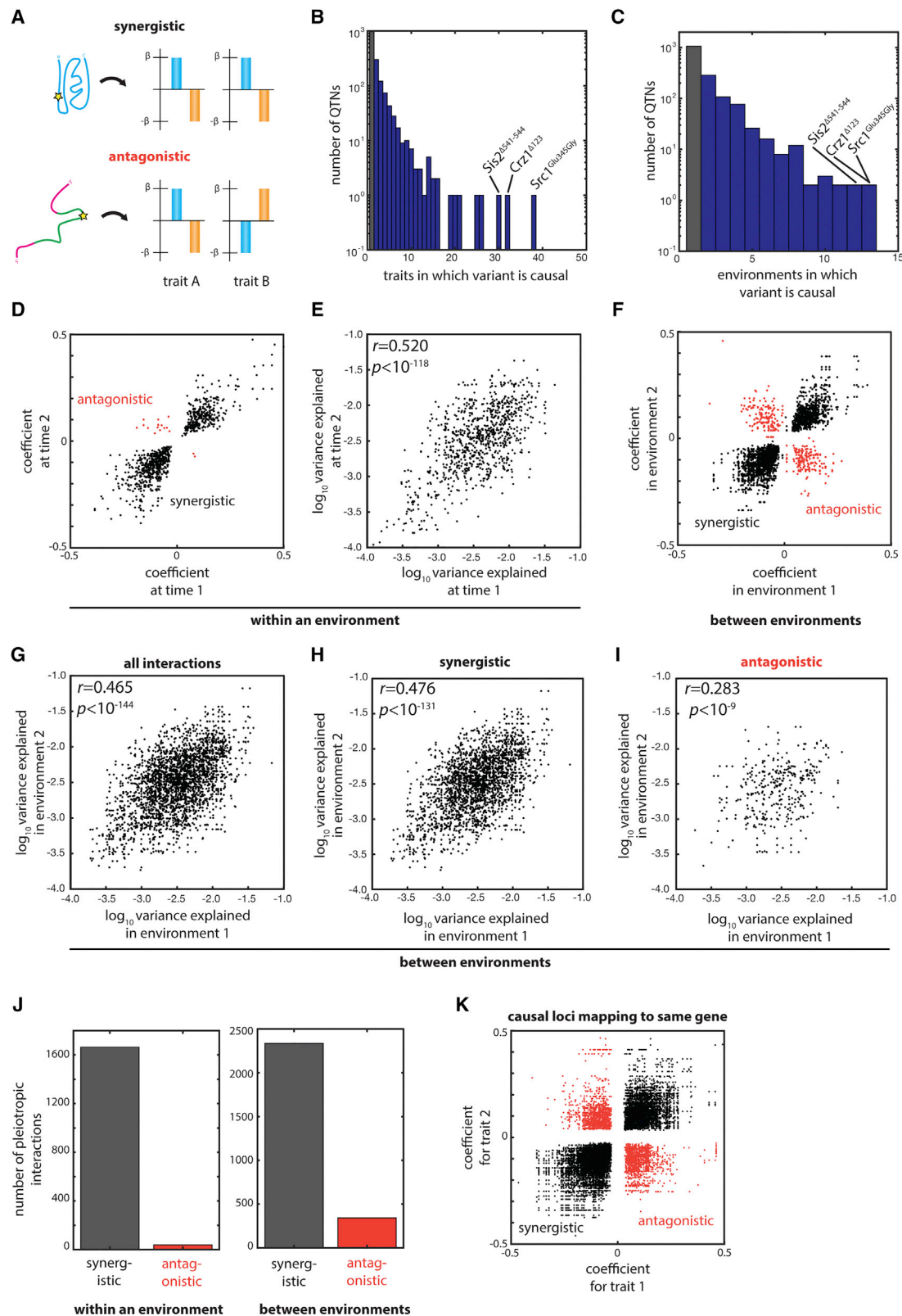


Figure 5. Abundant Synergistic and Antagonistic Interactions Resolved to Single Nucleotides

(A) Schematic of the phenotypic effects of synergistic and antagonistically pleiotropic variants.

(B) Histogram of number of traits for which each QTN was identified as causal.

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and diverse types of variation, including missense, regulatory, and synonymous variants (Figure S4A–S4C).

It is challenging to assign “true” pleiotropy on the basis of mapping studies that lack nucleotide resolution because it remains formally possible that distinct causal variants within the mapping window result in spurious apparent pleiotropy (Solo-vieff et al., 2013). Moreover, other variants within a gene could have opposing effects on a different phenotype, convoluting interpretation of the gene product’s biological role. Our results suggest that these concerns are well founded: many QTLs mapping to the same gene had distinct (often opposing) effects (Figure 5K).

Although variants often exhibited opposing effects across environments, the effect sizes of common QTNs across traits were significantly correlated, even for antagonistic gene-environment interactions (Figures 5E and 5G–5I). This implicates certain variants as key nodes in the mapping of genotype to phenotype, regardless of whether the allele is beneficial or detrimental in a given condition, and suggests that large fitness gains may be intrinsically tied to substantial fitness losses in other environments.

Highly Pleiotropic Variants Alter Disordered Regions of Signaling Hubs

The three most synergistically pleiotropic variants we identified were associated with key hubs in cellular information flow and exhibited coherent effects on phenotype. The most pleiotropic causal variant was a missense mutation (Glu345Gly) in the inner nuclear membrane protein Src1 (Figure S4D); the effect on growth was in the same direction across all 38 traits for which the locus affected phenotype. Src1 binds chromosomes at the telomere and sub-telomere and mediates transcription of sub-telomeric genes, including many associated with alternative carbon source and phosphate metabolism (Grund et al., 2008). Residue 345 of the protein is nuclear facing in both splice variants of Src1, implicating *SRC1*^{Glu345Gly} in telomere and sub-telomere binding and thus the transcriptional regulation of a wide array of genes.

Two other highly pleiotropic variants employed a common mechanism: contractions in disordered protein regions. The first, *CRZ1*^{ΔLeu123}, removed Leu123 from a poly-Q tract in the disordered N-terminal transcriptional activation domain of the transcription factor Crz1, an archetypal calcineurin target (from the same family as human NFAT) that mediates the transcriptional response to a broad range of insults (Matheos et al., 1997; Stathopoulos and Cyert, 1997). The *CRZ1*^{ΔLeu123} deletion exhibited a coherent phenotypic effect, with the same effect direction in all

32 traits (Figure 6A). We confirmed the widespread phenotypic importance of the *CRZ1* gene product by measuring the growth of a strain with a targeted deletion of the gene in the same battery of growth conditions used for mapping: the absence of *CRZ1* exerted a significant effect in 11 of the 15 environments tested ($p < 0.05$; Student’s two-tailed t test) (Figure S4). The *CRZ1*^{ΔLeu123} QTN was associated with substantial phenotypic changes in many traits (e.g., Figure 6C) but was in very close proximity to another segregating variant (*CRZ1*^{381G>A}) (Figure 6D). To confirm that the statistically identified QTN was responsible for changes in calcineurin-dependent signaling, we measured the activity of a 4×CDRE::LACZ reporter (Stathopoulos and Cyert, 1997) in representative haploid F₆ progeny of each ditype of these two loci. Signaling was impaired in segregants bearing the *CRZ1*^{ΔLeu123} QTN and not the neighboring synonymous variant (Figure 6E).

The other highly pleiotropic variant, *SIS2*^{Δ541–544}, removed four Asp residues from a long, unstructured acidic tract at the protein’s C terminus. Sis2 is a bifunctional protein that plays an enzymatic role in Coenzyme A biosynthesis, conferred by its N-terminal and central domains, and engages in a regulatory interaction with protein phosphatase 1 (PP1), conferred by the disordered C-terminal domain in which the causal amino acid contraction was located (De Nadal et al., 1998). This case is emblematic of the power of our mapping approach: resolving the causal variant to the C-terminal domain clarified the function of the variant and the mechanism of pleiotropy. Once again, the phenotypic effects of the variant were uniform, exhibiting a coherent effect in all 30 traits (Figure 6B). PP1 inhibition modulates the phosphorylation of Crz1, in turn controlling its nuclear localization and transcriptional activity (Ruiz et al., 2003). Indeed, deletion of *SIS2* was even more pleiotropic than the loss of *CRZ1*: a strain lacking *SIS2* exhibited a phenotype in all 15 mapped environments ($p < 0.05$; Student’s two-tailed t test) (Figure S4E). Our data suggest that the *SIS2*^{Δ541–544} variant impacts stress-responsive transcription by altering PP1 inhibition and in turn altered Crz1 activity. Full transcriptional activation of a 4×CDRE::LACZ reporter of Crz1 activity (Stathopoulos and Cyert, 1997) required *SIS2*, further supporting this model (Figure S4F).

The mechanistic convergence between these highly pleiotropic variants was two-fold and suggestive of general principles for pleiotropic molecular variation. First, both polymorphisms occurred in disordered regions (Figures 6F and 6G); these regions are increasingly recognized as key facilitators of regulatory interactions in cells (Uversky, 2014). Strong positive selection has recently been identified in disordered regions in *H. sapiens*

(C) Histogram of number of environments for which each QTN was identified as causal.

(D) Plot of discovered coefficients at each pair of time points for which a given QTN was identified as causal within a given environment.

(E) Plot of variance explained at each pair of time points for which a given QTN was identified as causal within a given environment. Correlation by Pearson’s r ; p value by Student’s t distribution.

(F) Plot of discovered coefficients in each pair of quantitative traits for which a given QTN was identified as causal between environments.

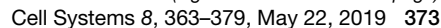
(G) Plot of variance explained in each pair of quantitative traits for which a given QTN was identified as causal between environments. Correlation by Pearson’s r ; p value by Student’s t distribution.

(H and I) Plot of variance explained in each pair of quantitative traits for which a given QTN was identified as (H) synergistically pleiotropic or (I) antagonistically pleiotropic between environments. Correlations by Pearson’s r ; p values by Student’s t distribution.

(J) Number of pleiotropic interactions identified as synergistic or antagonistic (left) within an environment or (right) between environments.

(K) Plot of discovered coefficients for each pair of quantitative traits in which a QTL could be attributed to a given QTG. Correlations by Pearson’s r .

See also Figure S3.



(Afanasyeva et al., 2018), suggesting that this kind of variant may be of broad importance. Second, both variants were in proteins related to a key regulatory process: signaling through the calcineurin and PP1 hubs, transduced by the transcription factor Crz1 (Ruiz et al., 2003; Thewes, 2014). Interestingly, both the polyglutamine tract of Crz1 and the acidic tract of Sis2, in which the two causal variants occurred, are highly polymorphic across *S. cerevisiae* strains and among related budding yeast species (Figures 6H and 6I) (Bergström et al., 2014). This, combined with the observation of extensive synergistic pleiotropy, suggests that these loci may be important sites for the generation of phenotypic heterogeneity.

Distribution of Fitness Effects of Extant Genetic Variation

The distributions of fitness effects (DFEs) of new and existing mutations are key in understanding the process, history, and potential of natural selection and evolution (Loewe and Hill, 2010; Orr, 2010). Diverse experimental approaches have measured the DFE of new mutations in bacteria and eukarya (Frenkel et al., 2014; Koufopanou et al., 2015; Robert et al., 2018). These experiments can be expected to sample, on average, mutations of larger effect than those that exist in the wild since the distributions at hand have not been subject to substantial purifying selection (other than to avoid lethality). Conversely, computational sequence-based approaches are subject to the limitation that we cannot accurately predict the effects of variants from sequence alone. Our mapping approach bridges this divide by combining direct measurement of the fitness effects of causal variants with uniform sampling of a real, naturally occurring distribution of fitness effects.

The DFE for a monogenic trait would have very few causal variants; at the other extreme of complexity, a trait that was truly “infinitesimal” (in Fisher’s sense of the limit of all segregating Mendelian factors; Barton et al., 2017) would present as many causal variants as there are extant variants, likely with a modal effect approaching zero (Figure 7A). Alternatively, a polygenic trait might instead exhibit a bimodal distribution with two distinct classes of variants: “causal” variants, which have a measurable effect on phenotype, and “nearly neutral” variants, whose effects are essentially negligible (Figure 7B). The shape of the DFE and the number of distinct classes of variants it contains have been the subject of much theoretical and experimental attention (Boucher et al., 2016; Orr, 2003; Rice et al., 2015).

We estimated the sensitivity of our mapping panel to causal loci of varying effect by performing *in silico* mapping of simulated hypothetical traits with underlying causal variants of known effect size (Figure 7C). We subsequently estimated the true un-

derlying distribution of fitness effects across the real quantitative traits we examined (Figure 7D). The apparent distribution we detected suggested that the true underlying distribution of fitness effects was bimodal: a discrete set of variants impacted each quantitative trait in a manner categorically distinct from the other, nearly neutral polymorphisms.

The shape of this inferred underlying distribution was similar for missense, synonymous, and extragenic variants (Figure 7E). It was also robust to the underlying distribution of effect sizes used to estimate sensitivity: we could accurately recover the distribution for normal, uniform, and monotonically decreasing effect-size distributions (Figures S5A–S5I). We observed no evidence of the Beavis effect, wherein genetic mapping can overestimate the effect of discovered QTLs because of closely linked causal variants (Beavis et al., 1991; King and Long, 2017); in fact, the coefficients were mildly deflated rather than overestimated. Our estimates of sensitivity were also robust to very large numbers of causal variants underlying the hypothetical traits used for calibration, ranging from 250 to 1,000 causal loci (of 12,054 segregating polymorphisms) per trait (Figures S5J–S5M).

The omnigenic model, as recently articulated by Pritchard and colleagues, is distinct from the limit of Fisher’s model (Boyle et al., 2017; Liu et al., 2018). It does not posit that all variants are causal *per se* but rather that sufficiently many “peripheral” genes contribute to phenotype that the majority of (or perhaps all) polymorphisms are in linkage disequilibrium with a causal variant. This results in apparent universal causality from the perspective of GWASs, as nearly all genomic regions make a meaningful predictive contribution. Our observations are broadly consistent with this model: 30.0% of segregating polymorphisms in our mapping panel are within 1 kb of a QTL marker for growth in 2% galactose, for example (Figure 7F), and 91.6% of segregating variants are within 1 kb of a QTL marker for at least one quantitative trait. The genome-wide distribution of QTLs we observe thus reconciles the statistical and theoretical architecture of the omnigenic model (Boyle et al., 2017; Liu et al., 2018) with our observation that not all segregating variants impact every quantitative trait.

DISCUSSION

Resolving large numbers of QTLs to individual causal variants allowed us to quantitatively address long-standing questions regarding the architecture of complex traits. Although coding variants had on average the greatest effect on phenotype, non-coding variants, including synonymous codons, had remarkably similar effect sizes. Indeed, the importance of such genetic

(C) Normalized colony sizes of F_6 diploid progeny with the indicated genotypes after 96 h of growth on media containing 2% glycerol. Error bars show SEM.

(D) Diagram indicating the locations of the $CRZ1^{ALeu123}$ QTN and the neighboring synonymous $CRZ1^{381G>A}$ variant in the $CRZ1$ gene.

(E) Normalized 4xCDRE::LACZ reporter activity for at least $n = 3$ F_6 haploid segregants of the indicated genotypes measured in biological triplicate. Bars show mean across the genotype; error bars show SEM. p value by Student’s two-tailed t test.

(F) Diagram of the Crz1 protein and predicted disorder from the Database of Disordered Protein Predictions (D2P2); the identified causal variant is indicated with a star.

(G) Diagram of the Sis2 protein and predicted disorder from D2P2; the identified causal variant is indicated with a star.

(H) Multiple sequence alignment of the Crz1^{111–145} region in the indicated *S. cerevisiae* strains and other yeast species.

(I) Multiple sequence alignment of the Sis2^{516–555} region in the indicated *S. cerevisiae* strains and other yeast species. See also Figure S4.

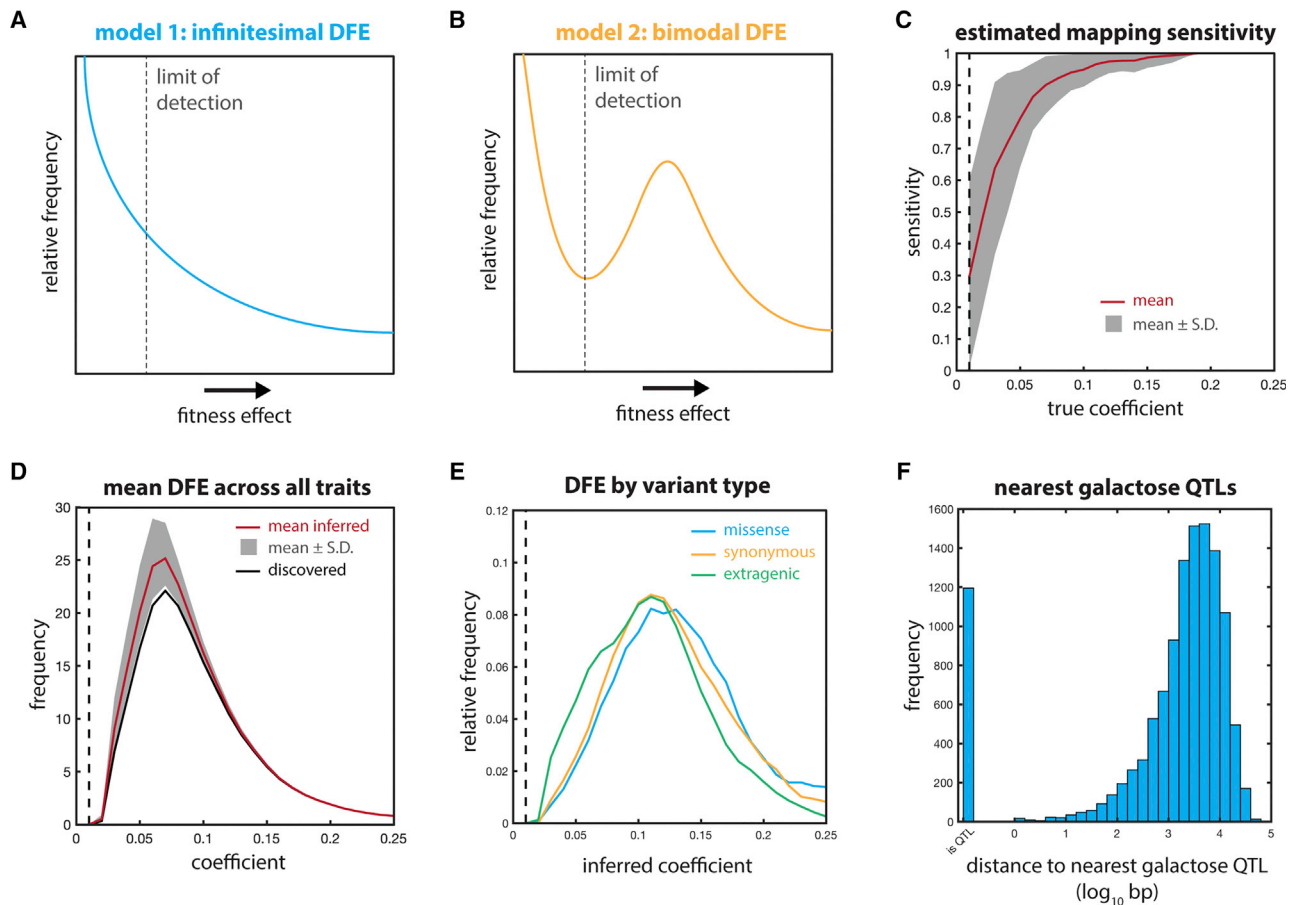


Figure 7. Inference of the Distribution of Fitness Effects of Extant Mutations

(A and B) Hypothetical true underlying fitness effect distributions under (A) infinitesimal and (B) bimodal models.

(C) Sensitivity of the mapping procedure to QTLs as a function of the true effect size for $n = 10$ simulations with 250 underlying causal variants.

(D) Inferred true underlying fitness effect distribution (per trait).

(E) Inferred average underlying fitness effect distributions based on discovered coefficients for missense (blue), synonymous (orange), and extragenic (green) variants.

(F) Distance of each segregating polymorphism in the mapping panel to the nearest identified QTL for growth on 2% galactose.

See also [Figure S5](#).

variation has been noted in organisms from *Salmonella* to humans (Kristofich et al., 2018; Supek et al., 2014). These observations suggest that the conventional wisdom regarding the prioritization of putative causal variants demands revision, as does the bioinformatic practice of assuming the neutrality, or near-neutrality, of “silent” variants relative to missense mutations. The importance of codon choice is evident from empirical studies in both bacteria and eukarya (Frumkin et al., 2018; Goodman et al., 2013; Tuller et al., 2010). Although the effects we observe can, in principle, be explained by many mechanisms, the pronounced biases in position and codon optimality suggest that the effects of changes in codon choice are primarily manifested early in translation. Nature, it seems, uses synonymous codons as a means of tuning translation rate, and presumably protein level, in a manner orthogonal to amino acid identity.

Variants outside of ORFs exhibited two likely modes of action that are not mutually exclusive, impacting the binding of transcription factors directly and changing the local structural

genomic context by altering the biophysical properties of the surrounding DNA. This idea was reinforced by our observation that causal variants were associated with open chromatin, in concordance with a privileged role for polymorphisms in transcriptionally active regions of the genome in determining phenotype (Roytman et al., 2018; Schaub et al., 2012; Trynka et al., 2013). Our results provide rigorous evidence in favor of the pragmatic assumptions underlying the integration of tissue-specific epigenomic and expression data with lower-resolution genetic mapping results (GTEx Consortium, 2017).

Antagonistic pleiotropy and antagonistic gene-environment interactions, phenomena of fundamental importance in understanding the patterns of emergence and fixation of novel molecular variation (Qian et al., 2012) as well as in designing safe and effective interventions for genetic medicine (Carter and Nguyen, 2011; Rodríguez et al., 2017), were strikingly common. Indeed, our observations likely represent a lower bound on the true extent of pleiotropy and gene-environment interactions, as

unresolved QTLs for one trait may be attributable to mapped QTNs for another. Two of the three most pleiotropic variants were coding variants in disordered regions of important cell-signaling proteins, Crz1 and Sis2, suggesting that the alteration of the interactions (Babu et al., 2011) mediated by disordered regions, long regarded as inert linkers, may be a general mechanism for the generation of phenotypic heterogeneity by minimal genetic change (Jakobson and Jarosz, 2018). Our findings offer fertile ground for the future study of specific examples of pleiotropy in a variety of genes, pathways, and environments and describe convolutions to the genotype-to-phenotype relationship that likely abound in the wild (Manuck and McCaffery, 2014; Via and Lande, 1987). It will also be fruitful in future to examine other phenotypes; the predominance of synergistic interactions observed here may not extend to traits beyond growth and proliferation.

Our results help to reconcile a vigorous debate (Cox, 2017; Liu, 2017; McMahon, 2017) regarding the omnigenic model: while seemingly equivalent to Fisher's "infinitesimal" model (which assumes infinitely many segregating causal alleles), an apparently omnigenic relationship between genotype and phenotype can often arise under realistic linkage disequilibrium without all segregating variants impacting a trait. We find that most quantitative traits likely comprise sufficiently many underlying causal loci as to appear omnigenic from the perspective of a typically (under)powered GWAS; nonetheless, enumerating these many contributors may be feasible.

Substantial opportunities remain for improvement in the construction of inbred mapping populations. Most notably, although our mapping panel has advantages in the detection and resolution of causal variants, it is not currently feasible to detect all epistatic interactions. Indeed, once *cis*- and *trans*-effects are considered, there exist in principle nearly one billion possible second-order terms, a problem of underdetermination exacerbated by our use of diploids. These higher-order interactions are known to be significant in many molecular contexts (Heck et al., 2006; Olson et al., 2014; Poelwijk et al., 2017) and in complex traits (Bloom et al., 2015; Forsberg et al., 2017) and are therefore important for a full understanding of heredity. The residual unexplained broad-sense heritability in our experiments was likely due primarily to these second- and higher-order effects. Much larger mapping panels will be required to address these questions while maintaining sufficiently low linkage disequilibrium to enable *in silico* fine-mapping.

Despite astounding advances in our ability to edit genes and even synthesize genomes, a conundrum persists: if one were to design an entire genome from scratch, what sequence should be chosen? Gene-level information is insufficient to answer this question if the goal is to optimize one or more quantitative traits. Despite the evident complexity of the problem, our results indicate that it will be possible to elucidate not only linear and dominant contributions to the genotype-to-phenotype relationship at the molecular level, as we have done, but also to thoroughly define higher-order contributions at nucleotide resolution. The use of appropriate experimental and statistical approaches in model organisms such as budding yeast will be critical in advancing our fundamental understanding of the highly complex relationships between genotypes and expressed traits. Indeed, fully defining such relationships in suffi-

cient detail may well prove intractable in humans and other metazoans without establishing their underlying architecture in models amenable to maximum-resolution quantitative genetics. Understanding the nature of quantitative traits will, in turn, become crucial as genome reading and writing become routine elements of not only scientific research but also medicine and industry.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- CONTACT FOR REAGENT AND RESOURCE SHARING
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 - Genetic Mapping
 - Gene Expression Analysis
 - Inference of Distribution of Fitness Effects
 - Phenotyping of the *S. cerevisiae* ORF Deletion Collection
 - Bioinformatic Analyses
 - Reporter of Calcineurin-Dependent Signaling
 - Computational Methods and Resources
- QUANTIFICATION AND STATISTICAL ANALYSIS
- DATA AND SOFTWARE AVAILABILITY

SUPPLEMENTAL INFORMATION

Supplemental Information can be found online at <https://doi.org/10.1016/j.cels.2019.04.002>.

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AUTHOR CONTRIBUTIONS

C.M.J. and D.F.J. conceptualized the study. C.M.J. performed the experiments and analysis. C.M.J. wrote the original draft of the manuscript. C.M.J. and D.F.J. edited and revised the manuscript. D.F.J. supervised the project.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, Peptides, and Recombinant Proteins		
SB3-14	Sigma	40772-50G
o-NPG	Bio Basic	ND0382
Deposited Data		
mRNA-seq data	This paper	GEO: GSE123702
F ₆ diploid genotype data	This paper	https://github.com/cjakobson/mapping
F ₆ diploid growth data	This paper	https://github.com/cjakobson/mapping
Experimental Models: Organisms/Strains		
RM11	NCYC	Liti et al., 2009
YJM975	NCYC	Liti et al., 2009
F ₆ haploid progeny	Jarosz Laboratory	She and Jarosz, 2018
F ₆ diploid progeny	This paper	N/A
BY4741 gene deletion/DAmP collection	Dharmacon/Thermo	N/A
Recombinant DNA		
4xCyDRE::LACZ reporter (pAMS366)	Cyert Laboratory	Stathopoulos and Cyert, 1997
Software and Algorithms		
Genetic mapping code	This paper	https://github.com/cjakobson/mapping
MATLAB R2016B	MathWorks	N/A
Kallisto	Pachter Lab	https://pachterlab.github.io/kallisto/download

CONTACT FOR REAGENT AND RESOURCE SHARING

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Daniel F. Jarosz (jarosz@stanford.edu).

EXPERIMENTAL MODEL AND SUBJECT DETAILS

The founding parental strains of the inbred cross are RM11a (*MATa ho::kanMX ura3Δ0 leu2Δ0*) and YJM975α (*MATα ho::hygMX uraΔ3::KanMX his3Δ1::NatMX*), both part of the *Saccharomyces* Genome Resequencing project and available from the NCYC. Also used were BY4741a (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0*), BY4741a Δ*CRZ1* (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 crz1Δ::kanMX*), BY4741a Δ*SIS2* (*MATa his3Δ1 leu2Δ0 met15Δ0 ura3Δ0 sis2Δ::kanMX*), and the complete BY4741a gene deletion and DAmP allele collection (Dharmacon/Thermo). The derivation and genotypes of the F₆ haploid progeny are described in detail elsewhere (She and Jarosz, 2018). The phased genotypes of the F₆ diploid progeny used here for genetic mapping are available at <https://github.com/cjakobson/mapping>.

METHOD DETAILS

Yeast Propagation and Phenotyping

Cells were revived from frozen stocks by pinning first to appropriate selective liquid medium (SD-Leu; SD+Hyg; SD-Leu+Hyg) using a Singer ROTOR robotic pinning instrument and thence to selective solid medium. Cells were transferred to various solid media (synthetic medium with yeast nitrogen base without amino acids, complete supplement mixture, and carbon source and drug/stressor as indicated; carbon source is 2% glucose if not indicated otherwise; see Table S2) for phenotyping using the same instrument and propagated for 144 hours. Growth was measured every 24 hours in 384-spot format by scanning the plates; colony size was quantified using the SGAtools suite (Wagih et al., 2013). Custom MATLAB code was used to normalize and Z-score colony sizes. Broad-sense heritability was estimated from biological replicates using a linear mixed-effects model with random effects (Bloom et al., 2013).

Cross Construction

We previously genotyped 1,125 F₆ haploid progeny of a cross between RM11-1a and YJM975 (She and Jarosz, 2018). Of these, we selected 384 Leu⁺Hyg[−] and 104 Leu[−]Hyg⁺ F₆ progeny (mixed Mat a/α) to generate the diploid panel used here. Each Leu⁺Hyg⁺ haploid

was mated to all 384 Leu⁺Hyg⁻ progeny on solid YPD medium for 24 hours, then transferred to solid diploid-selective medium (SD-Leu+Hyg), grown for 48 hours, then transferred to selective solid medium once more for 48 hours. The 384 Leu⁺Hyg⁻ progeny are approximately one-half MATa, so each mating yielded approximately 192 diploid progeny. Complementary mating type plates were merged so that each final plate of the collection contains two combined collections of diploids, with each collection sharing one parent. This is important to avoid conflating plate-to-plate variability effects, which are included in the regression model, with the effect of sharing a parent. Diploid selection plates were imaged before merging and positions that aberrantly contained cells in both source plates were excluded from the subsequent regression. The final collection contains 18,126 unique F₆ diploid strains stored in 52 384-well plates. Please contact jarosz@stanford.edu for information regarding obtaining the segregant panel.

Genetic Mapping

Diploid genotypes were constructed based on the haploid genotypes determined previously (She and Jarosz, 2018); phasing information was retained (although not used here) and loss of heterozygosity was neglected, as the diploids were propagated minimally before phenotyping. Variants present in the haploids had been called previously using SnpEff (Cingolani et al., 2012), and the coordinates shown are with reference to the strain S288C. The .vcf file associated with the segregating variants is available along with the full genotype matrix and code for the mapping procedure at <https://github.com/cjakobson/mapping>. 5' and 3' UTR variants were assigned on the basis of transcription start and end sites as reported in (Nagalakshmi et al., 2008). Extragenic QTNs were assigned to the nearest TSS.

The full 18,126-strain homozygous genotype matrix, comprising 12,054 markers, was regressed against normalized, Z-scored phenotype vectors for each quantitative trait using a forward stepwise selection routine. Strains lacking genotype or phenotype information were excluded from the regression. For the first iteration of the regression, only homozygous loci were included, with homozygous RM/RM loci taking a coefficient of +1 and homozygous YJM/YJM loci taking a coefficient of -1. Pseudogenotypes for plates and plate edges were included to account for positional effects. Markers were included in the model at a threshold of $p < 10^{-3}$ by *F*-test and removed at a threshold of $p < 10^{-2}$ by the same test. Following regression on the homozygous loci, the genotype matrix was expanded to 24,108 markers, with heterozygous loci taking coefficient +1 and homozygous loci taking coefficient 0 in the additional columns. Pseudogenotypes for plates and plate edges were again included. Forward stepwise selection was repeated using the same cutoffs with the final terms from the homozygous regression constituting the initial model.

Following marker selection by forward selection, causal variants with $p < 10^{-5}$ were fine-mapped by *in silico* allele swaps, as described previously (She and Jarosz, 2018). Briefly, all pairs of candidate loci within 10 markers of the putative marker from forward selection were compared by ANOVA, and a true causal variant (QTN) was declared when the null hypothesis that the marker in question was not causal could be rejected with respect to all other variants in the window. When multiple candidate QTNs could not be unambiguously distinguished but were all associated with a single gene, a QTG was declared instead. Otherwise, we recorded the marker as a QTL and did not include it in subsequent gene- or variant-specific analyses. Runtime for the entire mapping procedure (coarse and fine mapping) using MATLAB varied between 60 min and 120 min using hardware as described below, depending on the complexity of the trait. All mapped loci are annotated in [Data S1](#).

False discovery was controlled first at the level of QTLs by forward selection of the true genotype matrix against randomly permuted real data. An inclusion criterion of $p < 10^{-3}$ for stepwise selection and a final *p*-value cutoff of 10^{-5} yielded acceptable false discovery rates as shown in [Table S3](#). However, this approach does not give information on the accuracy of the fine-mapping procedure, nor on the recovery of the true underlying effect size of the causal variants. We therefore also conducted extensive simulations using 50 known ground-truth genetic architectures with 250 underlying linear causal variants and 25 underlying dominant causal variants, each with random, normally distributed effects, and using the true genotype structure of the panel ([Figure S1A](#)). Simulations were conducted with broad-sense heritability $H^2 \sim 0.85$, commensurate with that observed for the majority of traits we examined ([Figure S1C](#)). Our multivariate regression approach, followed by fine-mapping, performed favorably relative to a univariate method using Pearson's *r* to independently correlate each marker with phenotype ([Figure S1G](#)), which lacked the power to detect QTLs of small effect even in our very large mapping population. Hypothetical traits were used to tune the regression procedure, but the same training traits were not used to generate the validation results presented.

The causal genes identified across all traits formed a densely connected protein-protein interaction network (enrichment $p < 10^{-11}$ by STRING) and were enriched for 112 GO terms, including cellular metabolic process (FDR < 0.004), response to stress (FDR < 0.006), ion binding (FDR < 0.003), and response to nutrient levels (FDR < 0.03).

Gene Expression Analysis

Samples of exponentially growing (OD₆₀₀ ~ 0.5–1.0) 5 mL liquid cultures of homozygous diploid RM11a/α and YJM975a/α strains (in the media conditions indicated) were harvested in biological triplicate, snap-frozen, and stored at -80°C. RNA extraction, mRNA isolation by polyA enrichment, cDNA library preparation, and DNA sequencing were performed at the Beijing Genomics Institute, which returned ~20M clean, trimmed reads per sample deposited as GSE123702 at the NIH GEO. Reads were mapped and transcript abundances estimated using Kallisto (Bray et al., 2016) on the basis of the Ensemble *S. cerevisiae* cDNA reference. Estimated transcripts per million abundances for each ORF can be found in [Data S2](#).

Inference of Distribution of Fitness Effects

In order to estimate the sensitivity of the mapping panel and regression procedure to QTL of varying effect size, we generated hypothetical traits with $H^2 \sim 0.85$ and 250, 500, or 1000 underlying causal variants, with effect sizes drawn from normal, uniform, or quadratically decreasing distributions. The mapping procedure was conducted on $N = 10$ simulated traits for each underlying architecture using the true genotype structure of the panel, and sensitivity for each effect-size bin was estimated as the number of detected QTLs with effect sizes within the given range divided by the number of true underlying variants with those effect sizes. The true underlying effect size distribution was estimated as the number of discovered QTLs within a given effect size range for all the actual traits we considered divided by the estimated sensitivity for that bin based on the simulations.

Phenotyping of the *S. cerevisiae* ORF Deletion Collection

Strains bearing precise ORF deletions or hypomorphic DAmP alleles (Thermo Sci.) were revived from frozen stocks in liquid YPD medium, spotted to YPD solid medium, and grown on solid medium in 384-spot format (synthetic medium with yeast nitrogen base without amino acids, complete supplement mixture, and 2% ethanol) in biological duplicate for 48 hours. After 48 hours, growth was quantified as described above. Effect size of each deletion or hypomorphic allele was estimated as the Z-score of the mean colony size for that strain. The 100 core genes with the largest effect sizes were enriched for protein-protein interactions ($p < 0.03$; STRING database) and enriched for logically connected annotations (organelle organization, $p < 10^{-4}$; mitochondrial translation, $p < 10^{-3}$; mitochondrial matrix, $p < 0.05$; STRING database).

Bioinformatic Analyses

Chromosome and nucleotide positions of the extragenic variants were used to retrieve the surrounding nucleotide sequence of the S288C reference genome. Histone marks were averaged across all histones centered within 200 nt of the variant (Weiner et al., 2015). Similarly, TF occupancy from SwissRegulon was retrieved based on the chromosome and nucleotide positions of the extragenic variants relative to the S288C reference genome. Codon adaptation index was calculated as $f_{\text{codon } i} / \max(f_{\text{codons for that residue}})$ across the S288C reference genome, without adjustment for expression level. All statistical comparisons were conducted relative to all segregating variants of a given class within the mapping panel, not relative to simulated or uniform distributions.

Reporter of Calcineurin-Dependent Signaling

Calcineurin-dependent transcriptional activation was measured using the 4xCDRE::lacZ reporter plasmid pAMS366 (Stathopoulos and Cyert, 1997). The LacZ enzyme activity assay was conducted using standard methods. Briefly, cells with genotypes as indicated and bearing the episomal reporter were subcultured to $OD_{600} \sim 0.1$ in SD-CSM liquid medium with or without 100 mM Ca^{2+} , as indicated. After 4 hours of growth at $\sim 21^\circ\text{C}$, cell density at 600 nm was recorded and the cells were lysed for 90 min at 37°C using 10 g/L SB3-14 detergent in Z buffer (Miller, 1972). Following lysis, one-half of the reaction volume of 2 g/L chromogenic o-NPG substrate in prewarmed Z buffer was added and the mixture incubated at 37°C for 120 min. Samples were briefly centrifuged at $\sim 2,000 \times g$ to pellet insoluble cell debris and the supernatant transferred to a new plate. The LacZ hydrolysis product o-nitrophenol was monitored at 420 nm, and CDRE activity was calculated as A_{420}/A_{600} , adjusted for background scattering and nonspecific hydrolysis in lysate of cells lacking the reporter plasmid.

Computational Methods and Resources

Most computation was performed using MATLAB (MathWorks) on a MacBook Pro computer (2.7 GHz Intel Core i7; 16 GB RAM). Mapping was conducted using MATLAB on Sherlock nodes with 64 GB RAM. Also used were the SGATools (Wagih et al., 2013), clustalOmega (Sievers et al., 2011), D2P2 (Oates et al., 2013); Panther/GO (The Gene Ontology Consortium, 2017), SwissRegulon (Pachkov et al., 2007), Phyre2 (Kelley et al., 2015), SGRP (Bergström et al., 2014; Liti et al., 2009), and CellMap (Costanzo et al., 2016) web servers. The genotype and phenotype data are too large to include as supplementary files; all code required for mapping and validation, including segregant genotypes and all actual and simulated growth data, is deposited at <https://github.com/cjakobson/mapping>. Code used to generate the figures is available upon request to Daniel F. Jarosz (jarosz@stanford.edu).

QUANTIFICATION AND STATISTICAL ANALYSIS

Student's two-tailed *T*-test, *F*-test, binomial test, two-sample Kolmogorov–Smirnov test, and Fisher's exact test were performed in MATLAB. The Bonferroni correction was applied in the case of multiple testing. In mapping analyses, strains missing either genotype or phenotype information were excluded. Mean, median, standard deviation, and standard error of the mean are variously shown as indicated in the legends.

DATA AND SOFTWARE AVAILABILITY

All genetic mapping code is deposited at <https://github.com/cjakobson/mapping>. Other dependencies, including genotype and phenotype data too large to host on GitHub, can be downloaded from the link in the GitHub readme. mRNA-seq data are available at GEO: GSE123702 from the NIH GEO.